Appendix to Code availability is not enough: schemas for documenting and extending data analysis

Schema extraction for Team 5

For the many-analysts football study of Silberzahn et al. (2018), the workflow takes as input a pre-extracted methodology text for each team rather than the original report. The text used for Team 5 is quoted below, and Listing 1 gives the schema extracted from it, illustrating both the input and the output of the extraction stage.

> The two ratings of skin-tone were averaged and rescaled. The new variable was called avgrate01.
>
> Cases were excluded if they had missing values on skin-tone-rating, meanIAT or meanExp (listwise deletion) because we wanted to perform all analyses (including research question 2) on the same set of cases.
>
> The original response variable redCards is uninterpretable because the number of games a player has seen a given referee varies. Therefore we disaggregated the data (one game per row, redCards appear as 1s in the first n=redCards rows per player). This was possible because it does not matter in which of, for example, 3 games a player who received 1 red card in 3 games received the red card. It is sufficient that this player has three observations (three rows) associated with him, one of them indicating a red card. We used the binomial error distribution because – after disaggregation – our response variable specifies the occurence of an event in a single game, coded 0 and 1.
>
> We estimated generalized linear mixed models (function glmer in R package lme4, Version 1.1-7; Bates, 2010; Bates, Maechler, Bolker, & Walker, 2014). The crowd-storming data are different from standard multilevel data (e.g., where employees are members of only one team), inasmuch as they are not nested but cross-classified – player A can have multiple games with the referee A, but player B can have multiple games with the same referee A.

Our model adds a random effect of playerShort, refNum, and skin-tone across referees' countries of origin. We did not use any covariates, even though reviewers of our approach suggested that we should do so. As already noted in the project description by Silberzahn, Martin, Uhlmann, & Nosek, the data cannot be used for causal inference.

Listing 1: The schema extracted from the Team 5 methodology text quoted above. Two steps are returned with medium confidence, each carrying the clarification question the model would put to the analyst.

```yaml
meta:
  type: schema
nodes:
- id: create_skin_tone_variable
  objective: how to operationalize player skin tone from multiple rater assessments
  decision: average the two ratings of skin-tone and rescale to create avgrate01
  rationale: to combine multiple rater assessments into a single composite measure
  inputs: [rater1, rater2]
  outputs: [avgrate01]
  confidence: high
  clarification_question: null

- id: handle_missing_data
  objective: how to handle cases with missing values on key variables
  decision: apply listwise deletion excluding cases with missing values on
   ↪ skin-tone-rating, meanIAT, or meanExp
  rationale: to perform all analyses including research question 2 on the same set of
   ↪ cases
  inputs: [avgrate01, meanIAT, meanExp]
  outputs: [df_complete]
  confidence: high
  clarification_question: null

- id: disaggregate_response_variable
  objective: how to handle the response variable given varying number of games per
   ↪ player-referee dyad
  decision: disaggregate data to one game per row where redCards appear as 1s in the
   ↪ first n=redCards rows per player
  rationale: the original response variable redCards is uninterpretable because the
   ↪ number of games a player has seen a given referee varies
  inputs: [df_complete, redCards, games, playerShort, refNum]
  outputs: [df_disaggregated, red_card_binary]
  confidence: medium
  clarification_question: Which column indicates the total number of games per
   ↪ player-referee combination for disaggregation?

- id: select_error_distribution
```

```
  objective: which error distribution to use for modeling the response variable
  decision: use binomial error distribution
  rationale: after disaggregation the response variable specifies the occurrence of
   ↪ an event in a single game coded 0 and 1
  inputs: [red_card_binary]
  outputs: [distribution_spec]
  confidence: high
  clarification_question: null

- id: specify_random_effects_structure
  objective: how to account for the non-nested cross-classified data structure
  decision: add random effects of playerShort, refNum, and skin-tone across referees
   ↪ countries of origin
  rationale: the crowdstorming data are cross-classified where players can have
   ↪ multiple games with multiple referees
  inputs: [playerShort, refNum, avgrate01, refCountry]
  outputs: [ranef_spec]
  confidence: medium
  clarification_question: Is refCountry the correct column representing referees
   ↪ countries of origin for the random slope specification?

- id: select_covariates
  objective: whether and which covariates to include in the model
  decision: do not include any covariates
  rationale: the data cannot be used for causal inference as noted in the project
   ↪ description
  inputs: [df_disaggregated]
  outputs: [covariate_spec]
  confidence: high
  clarification_question: null

- id: fit_glmm
  objective: which modeling framework to use for the analysis
  decision: estimate generalized linear mixed models using glmer function from lme4
   ↪ package
  rationale: appropriate for binary outcome with cross-classified random effects
   ↪ structure
  inputs: [df_disaggregated, red_card_binary, avgrate01, ranef_spec,
   ↪ distribution_spec, covariate_spec]
  outputs: [glmm_model]
  confidence: high
  clarification_question: null
```

Listing 2 gives the script generated and validated from that schema, completing the pipeline from methodology text to executable code for this team. As in the main paper, the script is annotated with the schema step each block implements. Although the schema specifies `glmer()` from `lme4`, the script fits the model with `glmmTMB()`, reflecting the substitution of

faster model-fitting functions described in the main paper.

Listing 2: The R script generated and validated from the schema in Listing 1.

```r
# Load required libraries
library(readr)
library(dplyr)
library(glmmTMB)

# Load the data
df <- read_csv("data/football.csv")

# Step 1: create_skin_tone_variable
# Average the two ratings of skin-tone and rescale to create avgrate01
# Input: rater1, rater2
# Output: avgrate01
df <- df %>%
  mutate(avgrate01 = (rater1 + rater2) / 2)

# Rescale to 0-1 range
avgrate_min <- min(df$avgrate01, na.rm = TRUE)
avgrate_max <- max(df$avgrate01, na.rm = TRUE)
df <- df %>%
  mutate(avgrate01 = (avgrate01 - avgrate_min) / (avgrate_max - avgrate_min))

# Step 2: handle_missing_data
# Apply listwise deletion for cases missing skin-tone-rating, meanIAT, or meanExp
# Input: avgrate01, meanIAT, meanExp
# Output: df_complete
df_complete <- df %>%
  filter(!is.na(avgrate01) & !is.na(meanIAT) & !is.na(meanExp))

# Step 3: disaggregate_response_variable
# Disaggregate data to one game per row with redCards appearing as 1s in first n rows
  ↪ per player
# Input: df_complete, redCards, games
# Output: df_disaggregated, red_card_binary
df_disaggregated <- df_complete %>%
  filter(games > 0) %>%
  rowwise() %>%
  do({
    row_data <- .
    n_games <- row_data$games
    n_red_cards <- min(row_data$redCards, n_games)  # Ensure red cards don't exceed
  ↪ games

    # Create expanded rows
    expanded <- data.frame(
```

```
      playerShort = rep(row_data$playerShort, n_games),
      refNum = rep(row_data$refNum, n_games),
      refCountry = rep(row_data$refCountry, n_games),
      avgrate01 = rep(row_data$avgrate01, n_games),
      meanIAT = rep(row_data$meanIAT, n_games),
      meanExp = rep(row_data$meanExp, n_games),
      red_card_binary = c(rep(1, n_red_cards), rep(0, n_games - n_red_cards)),
      stringsAsFactors = FALSE
    )
    expanded
  }) %>%
  ungroup()

# Ensure factor variables for random effects
df_disaggregated <- df_disaggregated %>%
  mutate(
    playerShort = as.factor(playerShort),
    refNum = as.factor(refNum),
    refCountry = as.factor(refCountry)
  )

# Step 4: select_error_distribution
# Use binomial error distribution
# Input: red_card_binary
# Output: distribution_spec
distribution_spec <- binomial(link = "logit")

# Step 5: specify_random_effects_structure
# Add random effects for playerShort, refNum, and skin-tone across referees countries
   ↪  of origin
# Input: df_disaggregated, playerShort, refNum, avgrate01, refCountry
# Output: ranef_spec
ranef_spec <- "(1 | playerShort) + (1 | refNum) + (0 + avgrate01 | refCountry)"

# Step 6: select_covariates
# Do not use any covariates
# Input: df_disaggregated
# Output: covariate_spec
covariate_spec <- NULL

# Step 7: fit_glmm
# Estimate generalized linear mixed models using glmer in R package lme4
# Input: df_disaggregated, red_card_binary, avgrate01, ranef_spec, distribution_spec,
   ↪  covariate_spec
# Output: glmm_model

# Build the formula
# Fixed effect: avgrate01 (skin tone predictor)
```

```r
# Random effects: as specified in ranef_spec
# No covariates as specified in covariate_spec

formula_string <- paste("red_card_binary ~ avgrate01 +", ranef_spec)
model_formula <- as.formula(formula_string)

# Fit the GLMM
glmm_model <- glmmTMB(
  formula = model_formula,
  data = df_disaggregated,
  family = distribution_spec,
)

# Print model summary
summary(glmm_model)
```

Built-in prompts

The prompts that `tines` uses at each stage of the workflow are reproduced below, with placeholders in square brackets.

Prompt: extracting the schema from the text

```
You are an expert methodologist and data pipeline architect. I will
provide a text describing a multiverse analysis and a summary of the
actual dataset being used.

=== YOUR TASK ===

Extract a chronological list of methodological decisions (nodes) AND
map the exact data flow (inputs/outputs) for each node
simultaneously.

=== RULES ===

1. THEORY EXTRACTION: For each node, extract:

- 'id': A unique snake_case identifier.

- 'objective': MUST be framed as an open methodological goal that
invites multiple possible approaches. It must NOT describe the final
choice.
```

```
- 'decision': A 'decision' is strictly a POSITIVE methodological
choice that has potential theoretical alternatives, chosen to resolve
the 'objective'.

- 'rationale': WHY that decision was made, extracted from the text.

2. DATA MAPPING: Assign 'inputs' (EXACT column names from the dataset
OR outputs from previous nodes) and 'outputs' (invented snake_case
objects like 'df_clean' or 'ranef_spec').

3. ANTI-ABSTRACTION (CRITICAL): If the text lists specific variables
(e.g., 'centered rater, meanIAT'), DO NOT summarize them away. You
MUST capture those specific variables in the 'inputs' array.

4. INLINE ARRAYS: Format arrays strictly on one line: `inputs: [var1,
var2]`.

5. CONFIDENCE & CLARIFICATION: Rate your mapping confidence (HIGH,
MEDIUM, LOW). If the text abstracts a step and you cannot confidently
match it to specific dataset columns, set confidence to LOW and
autogenerate a 'clarification_question' asking the user which exact
columns to use. If HIGH, output 'null'.

6. Output ONLY valid YAML without markdown formatting.

=== REQUIRED YAML STRUCTURE EXAMPLE ===

meta: type: schema nodes: - objective: variables are in different
scales decision: apply min-max scaling to each variable rationale: to
put them on the same scale for combination id: step-scaling
confidence: high - objective: combine the school variables into one
dimension decision: average exp sch and avg sch rationale: the most
intuitive way id: step-education confidence: low

=== DATASET SUMMARY ===

NA

=== METHODOLOGY TEXT ===

[insert text here]
```

Prompt: generating code from the schema

```
You are an expert R programmer. Attached is a text document
containing a SCHEMA that defines a data processing pipeline.

No data file is specified - if data loading is required, use a
sensible placeholder or note it in a comment. Your task is to write a
complete, working R script that implements this pipeline
step-by-step. Use modern R practices (like dplyr or base pipe) and
ensure variables flow correctly from one step to the next. Output
ONLY the complete R script. Do not start with markdown formatting
blocks (like ```R) or backticks.
```

Prompt: validating the generated code

```
You are an expert R developer and automated debugging agent. The user
has provided a data file located at the relative path: `data/[insert
the path to the data]'. Your purpose is to receive broken R scripts,
diagnose the execution error, and return the corrected script.

CRITICAL INSTRUCTIONS: 1. Fix the bug while preserving the original
intent and logic of the script.  2. If the error is a missing
package, add the necessary `library()` call at the top.  3. If the
error is a missing variable, ensure it is properly initialized before
use.  4. Do NOT hallucinate new data files or external dependencies
unless explicitly provided in the context.

STRICT OUTPUT CONSTRAINTS: - You must output ONLY valid, fully
executable R code.  - Absolutely NO markdown formatting. Do NOT wrap
your response in ```R or ```.  - Absolutely NO conversational filler,
greetings, explanations, or comments about what you fixed.  - The
first character of your response must be R code, and the last
character must be R code.
```

Prompt: generating alternatives for a specific step

```
You are an expert Data Analyst and Methodologist. You are reviewing
an analysis schema to identify "Forking Paths" -- alternative
analytical choices that are equally valid but distinct from the
current approach.

=== DEFINITIONS ===
```

```
The schema provided to you consists of steps with:

- **OBJECTIVE**: The goal of the step (What needs to be done).

- **DECISION**: The specific implementation chosen (How it is done).

- **RATIONALE**: The reasoning behind that decision.

- **ID**: The unique identifier for the step (kebab-case).

=== TASK ===

Focus specifically on the step tagged: "[insert step number here]".
Your goal is to generate 3 distinct, valid alternatives for this
step.

For each alternative: 1. **Keep the same OBJECTIVE** (the goal
remains constant).

2. **Change the DECISION** to a different but methodologically sound
approach.

3. **Provide a new RATIONALE** explaining why this alternative is
valid.

4. **Create a new ID** that reflects the new decision (must be
kebab-case).

=== OUTPUT FORMAT ===

Please output the result in **strictly valid YML format**.

**Crucial Formatting Rules:**

1. Include a `meta` section at the top with `type: alternatives` and
`branch: multi`.

2. Output strictly valid YML. All text values (decision, rationale)
must be enclosed in double quotes ("). Do not use block styles (| or
>). Do not wrap lines or insert \n characters within the quotes; keep
the text as a single continuous string.
```

```
3. Do not include markdown code fences (like ```yml) or
conversational text. Just the raw YML.

=== REQUIRED YML STRUCTURE EXAMPLE ===

meta: type: alternatives branch: multi nodes: - overrides: [insert
step number here] alternatives: - id: step-new-method-name decision:
"Description of the new decision..."  rationale: "This is the
reasoning for why this alternative is valid."  - id:
step-another-method ...
```

References

Silberzahn, R., Uhlmann, E. L., Martin, D. P., Anselmi, P., Aust, F., Awtrey, E., Bahník, Š., Bai, F., Bannard, C., Bonnier, E., Carlsson, R., Cheung, F., Christensen, G., Clay, R., Craig, M. A., Dalla Rosa, A., Dam, L., Evans, M. H., Flores Cervantes, I., … Nosek, B. A. (2018). Many Analysts, One Data Set: Making Transparent How Variations in Analytic Choices Affect Results. *Advances in Methods and Practices in Psychological Science*, *1*(3), 337–356. https://doi.org/10.1177/2515245917747646